Fast Task-Specific Region Merging for SAR Image Segmentation

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***Abstract*— In existing superpixel-wise segmentation algorithms, superpixel generation most often is an isolated preprocessing step. The segmentation performance is determined to a certain extent by the accuracy of superpixels. However, it is still a challenge to develop a stable superpixel generation method. In this article, we attempt to incorporate the superpixel generation and merging steps into an end-to-end trainable deep network. First, we employ a recently proposed differentiable superpixel generation method to over-segment the single-polarization synthetic aperture radar (SAR) image. It outputs the statistical likelihood that each pixel belongs to different superpixels. In superpixel merging part, as one of our main contributions, we propose a superpixel-wise statistical dissimilarity measure method for converting the soft superpixels set into a self-connected weighted graph. More importantly, inspired by the concept of the number of walks in graph theory, we define the k-order connectivity of each vertex. This definition can intelligently indicate the potential soft cluster centers and class assignments in graph. This merging method is differentiable, computationally simple, and free of empirical parameters. The superpixel generation and merging phases can be implemented under a unified deep network. The benefit is that our method can iteratively adjust the shapes of the superpixels according to the boundaries and segmentation results during training, until the satisfactory segmentation results are captured. Experimental results on real SAR images demonstrate that the segmentation precision of our proposed method is superior to other state-of-the-art methods in terms of precision and computational efficiency.**

***Index Terms*— k-order connectivity, number of walks, segmentation, superpixel merging, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) can continuously monitor specific areas without constraints of weather and light. SAR image segmentation focuses at dividing the SAR images into different regions according to the pixels' features, which plays an important role in flood detection [1], sea ice estimation [2], agricultural output estimation [3], and other applications.

The current segmentation algorithms of SAR images can be divided into two branches. The first one takes pixels as processing units, and the other takes superpixels as processing units. Pixel-level segmentation methods generally first use gray-level co-occurrence matrix (GLCM) [4], wavelet transform [5], or other methods to extract the features and employ the classification or clustering methods (e.g., random forest (RF) [6], AdaBoost [7]) to segment the pixels into multiple regions. The common shortcoming of this type of method is high computational complexity and susceptive to speckle [8]. In order to accelerate the image segmentation, Shang *et al.* [9] selected a subset of special "key" pixels and performed segmentation on only these key pixels. Then, the remaining non-key pixels can be rapidly segmented guided by the key pixels. Nogueira *et al.* [10] introduced the statistical properties of log-cumulant estimates of SAR data to perform unsupervised SAR image segmentation. It generates an initial classification using the Gaussian mixture and the probability density of the G^0 parameters to provide the training samples for the classifiers. In order to effectively suppress the influence of speckle, Zhao *et al.* [11] proposed a fuzzy clustering algorithm based on gamma distribution for SAR image segmentation, where Stirling equation was used to approach the gamma function. Rocha Neto *et al.* [12] also presented a level set-based SAR image segmentation algorithm. This method embodies much of the data statistics without assuming any prior model for it so that it could deal satisfactorily with speckle on the intensity and amplitude SAR images. Recently, fast-growing deep learning techniques, such as convolutional neural network (CNN) [13] or recurrent neural network (RNN) [14], have indicated excellent feature expression and model fitting performance. Encouraged by this, researchers have proposed some deep SAR image segmentation methods. For example, Jiang *et al.* [15] presented a hierarchical fusion CNN model for SAR image segmentation. At each convolutional layer, it sets several different-sized receptive fields to accurately capture the structural and regional information. It is worth pointing out that these deep learning-based methods generally require sufficient training data, which hinders their further application in practice.

Manuscript received September 12, 2021; revised December 7, 2021; accepted January 4, 2022. Date of publication January 7, 2022; date of current version March 17, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62171015, Grant 62171016, Grant 61871413, and Grant 61801015; and in part by the Fundamental Research Funds for the Central Universities under Grant XK2020-03. *(Corresponding author: Fan Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3141125

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The superpixels are groups of similar neighboring pixels, which are perceptually meaningful and representationally efficient segments [16]. Because superpixels are much less than pixels, superpixel-level segmentation methods are generally computationally efficient and have received considerable attention. These methods usually first over-segment the whole image into a set of superpixels. For example, Zhang *et al.* [17] utilized minimum spanning tree (MST) to generate high quality multiscale superpixels with low computational costs. Arisoy and Kayabol [18] took pixel amplitudes and coordinates as features and resorted to finite mixture models [19] to cluster the pixels into superpixels. After that, the superpixels are classified or clustered to different landcovers such as urban, land. For example, Tian *et al.* [20] proposed a superpixel-wise fuzzy c-means (FCM) clustering approach, where the influence of neighboring and similar superpixels was incorporated into FCM to improve the segmentation performance. In order to suppress noise in SAR images and also ensure that small targets are not destroyed, Shang *et al.* [21] first used the image edges as constraints to conduct Gaussian smoothing. Then, K-means clustering and Markov random field (MRF) were conducted on the smoothing results sequentially to predict the pixel labels. Akyilmaz and Leloglu [22] modified the distance measure in simple linear iterative clustering (SLIC) [23] to similarity ratio metric based on the likelihood ratio. In the latter phase, the generated superpixels were clustered into larger segments using the same similarity metric. Carvalho *et al.* [24] used statistical region growing and hierarchical merging strategy to segment SAR images. Similarly, Shui and Zhang [25] utilized multiscale Bhattacharyya distance [26] to generate initial over-segmentation and then the region-merging process was performed with a merging cost in consideration of statistical similarity measure and the relative common boundary length penalty (RCBLP). Xiang *et al.* [27] further improved the computation efficiency of the superpixel merging process. It adopted the framework of statistical region merging (SRM), which had a lower computation cost and did not need to set any parameters in advance. Fan *et al.* [28] introduced texture pattern attention into region merging for a single-polarization SAR image segmentation. Zhang *et al.* [29] proposed a parallel region-merging criterion based on the region adjacency graph (RAG) of an initial partition and the nearest neighbor graph of the RAG. Shang *et al.* [30] employed the superpixel boundary to carry the strong and weak edge information of the image so that the edge information could participate in the final superpixel merging stage. Experimental results showed that it effectively improved the accuracy and smoothness of image edge segmentation. Inspired by the great development of deep learning in SAR image interpretation, some researchers also have sought to modify deep networks to incorporate SAR image segmentation. Ma *et al.* [31] presented an attention deep graph convolution network (GCN) [32] for superpixel-wise SAR segmentation.

It should be noted that most of the above-mentioned methods still regard the superpixel generation and region merging as two independent parts. As shown in Fig. 1(a), the superpixel generation is more like an isolated preprocessing step. Due to speckle noise and complex geometric distortions within SAR images, there are always some initial partitions poorly adhering the real boundaries. The subsequent merging cannot adjust the shapes of these worry superpixels and these confusions in over-segmentation will directly lead to the segmentation errors. In other words, the performance of segmentation methods is determined by the accuracy of superpixels to a certain extent. To solve this problem, we attempt to incorporate SAR superpixel generation and merging steps into an end-to-end trainable deep network. In this case, we can synchronously update the shapes of superpixels and merging strategies until the optimal segmentation results are recorded.

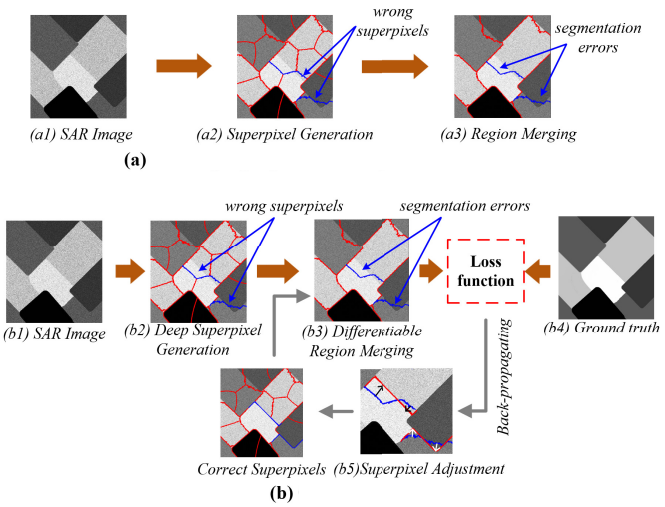


Fig. 1. Comparison of algorithmic schemes: (a) existing superpixel-wise methods. Superpixel generation and merging are regarded as two independent parts, where worry superpixels will directly lead to the incorrect segmentation and (b) proposed end-to-end superpixel-wise segmentation method. It can iteratively adjust the shapes of superpixels according to the segmentation errors until the optimal segmentation results are captured.

The main hurdle for incorporating them into a united network is that most of the existing initial over-segmentation and region-merging methods are both nondifferentiable. This results in that gradient descent will not be able to make a progress in updating the weights. In the superpixel generation stage, we adopt the superpixel sampling network proposed by Yang *et al.* [33]. This method revised the iterative k-means clustering algorithm to make it differentiable so that the superpixel estimating procedure can be implemented using a fully convolutional network. It outputs a soft association map, whose elements are continuous variables and represent the affiliation between this superpixel and all pixels.

In the superpixel merging phase, classical deep CNN or clustering methods cannot directly merge this kind of soft superpixels. To address this problem, we present a definition of k-order connectivity for each superpixel, by extending the concept of the number of walks for the unweighted graph to that for the weighted self-connected graph. To be specific, a graph can be regarded as a railway network where each vertex is a station. The k-order connectivity of a station is calculated by the entries of graph's similarity matrix, which represents how many other stations are connected to it and how busy this station is. Through a simple comparison between k-order connectivity values of different vertexes,

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we can determine which vertexes are soft cluster centers. This clustering method is computationally simple and differentiable and has no manually set parameters, meaning that it can be implemented using a deep network.

Fig. 1 illustrates the difference between our method and exciting superpixel-wise segmentation methods. As shown in Fig. 1(b), the deep superpixel generation network and our proposed superpixel-wise grouping method can be implemented under a deep network framework. For the worry superpixels, our method can calculate the loss values and then update the parameters of two part to adjust the superpixels, until all superpixels can correctly adhere the boundaries.

Overall, in comparison with other state-of-the-art methods, our superpixel-wise segmentation method has made the following contributions.

1) Inspired by the concept of the number of walks in graph theory, we present a definition of k-order connectivity for superpixels. Based on this, we propose a computationally simple, differentiable, and free of merging parameters clustering method. Before this, rare superpixel-wise grouping methods could be implemented using deep networks.
2) We propose a superpixel-wise statistical dissimilarity measure method for converting the soft superpixels set into a self-connected weighted graph.
3) Based on the above work, we construct an end-to-end superpixel-wise segmentation network to combine superpixel generation and merging. Two sets of parameters can be updated simultaneously to capture the globally optimal solution. In other words, in the process of segmentation, the shapes of superpixels are no longer invariable, but can be changed with the segmentation result.

The rest of this article is organized as follows. Deep superpixel generation algorithm is described in Section II. Section IV presents the fast superpixel merging algorithm based on k-order connectivity. Experimental results and the discussion are presented in Section V. Section VI presents the conclusion.

II. Semisupervised Deep Superpixel Generation Network

We illustrate a schematic of the proposed framework in Fig. 2. First, when an input SAR image is given, it is over-segmented by deep superpixel generation network. It outputs the likelihood that each pixel belongs to different superpixels, i.e., soft association map. Then, we propose a superpixel-wise statistical dissimilarity measure method, which aims at converting the soft superpixels set into a self-connected weighted graph. Finally, and most importantly, we define the k-order connectivity of each vertex. In the merging step, the superpixels are intelligently grouped based their k-order connectivity to acquire the segmentation. During the training stage, the superpixel generation and merging parameters can be dynamically refined based on the label difference between the segmentation output and superpixels' labels. More details are given in Sections II–IV.

Most of the existing over-segmentation algorithms (e.g., Watershed [34], TurboPixels [35]) are nondifferentiable, and

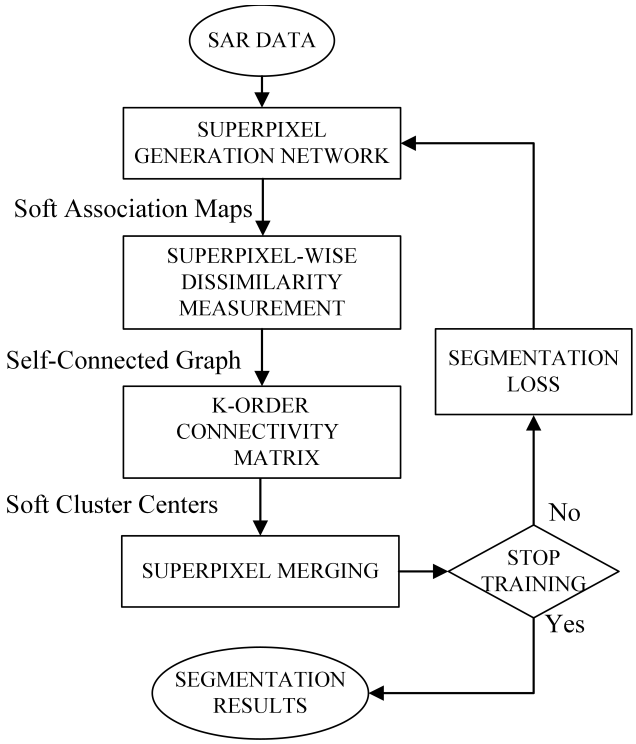


Fig. 2. Framework of our algorithm.

hence they cannot be implemented using deep networks. To address this problem, Jampani *et al.* [36] proposed a differentiable SLIC method for optical images by replacing the nearest neighbor operation with soft association map. In this framework, CNN becomes a feature extractor and the differentiable SLIC is applied to cluster the homogeneous pixels into the same superpixels. For each input SAR image, this clustering method needs to iteratively update the cluster centers until either convergence or for a fixed number of iterations. Hence, this model has relative heavy computation cost. Yang *et al.* [33] further improved Jampani's model. They directly employ an encoder–decoder deep convolutional network to estimate the superpixels. Experiments show that it consistently outperforms state-of-the-art method with low computational complexity. In this article, we introduce this deep superpixel generation approach to over-segment the whole SAR images into soft superpixels. Considering the difference between optical images and SAR images, we have made certain modifications to this model. Here, we will explain the process in detail.

Given an SAR image $\mathbf{I} \propto \mathbb{R}^{H\times W}$ with size $H \times W$, we first over-segment $\mathbf{I}$ into M grid cells with size of $h \times w$. The used encoder–decoder deep network converts $\mathbf{I}$ into a soft association matrix $\mathbf{Q} \propto \mathbb{R}^{H\times W\times M}$. The entry $q_{i,j}^{m}$ of $\mathbf{Q}$ denotes the probability of assigning pixel $f_{i,j}$ to the mth superpixel. The encoder employs multiple convolutional layers to extract the high-level feature maps from SAR images and the decoder then projects these feature maps to a soft association map using deconvolutional layers.

Superpixels are composed of spatially adjacent and similar pixels, so the over-segmentation process needs to consider both the intensity and position information of the pixels. Let $a_{i,j}$

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and $p_{i,j} \propto \mathbb{R}^{1\times 2}$ represent the intensity and coordinates of pixel $f_{i,j}$. The properties of all pixels can be written as a property matrix $\Phi \propto \mathbb{R}^{H\times W\times 3}$. Deep superpixel generation network takes Φ as input and outputs $\mathbf{Q}$. Then, we can obtain the intensity and location of superpixel center m, represented $\hat{a}_m$ and $\hat{p}_{\boldsymbol{m}}$

$$\hat{a}_m = \frac{\sum_{(i,j)\in \mathrm{I}} a_{i,j}\cdot q_{i,j}^m}{\sum_{(i,j)\in \mathrm{I}} q_{i,j}^m}, \quad \hat{p}_m = \frac{\sum_{(i,j)\in \mathrm{I}} p_{i,j}\cdot q_{i,j}^m}{\sum_{(i,j)\in \mathrm{I}} q_{i,j}^m} \tag{1}$$

where $\sum_{(i,j)\in \mathrm{I}} q_{i,j}^m$ is the normalization constant. Mathematically, $\hat{a}_m$ and $\hat{p}_{\boldsymbol{m}}$ are the weighted mean of all pixels' intensity values and locations. For convenience, we can normalize matrix $\mathbf{Q}$ to make $\sum_{(i,j)\in \mathrm{I}} q_{i,j}^m = 1$ and above superpixel center calculation process can be simplified as

$$\begin{bmatrix} \hat{a}_1 & \hat{p}_1 \\ \vdots & \vdots \\ \hat{a}_M & \hat{p}_M \end{bmatrix} = \tilde{\mathbf{Q}}^T \Phi \tag{2}$$

where $\tilde{\mathbf{Q}}$ is the normalized $\mathbf{Q}$ meeting $\sum_{(i,j)\in \mathrm{I}} q_{i,j}^m = 1$, and T denotes matrix transpose operation.

Then, given $\mathbf{Q}$ and superpixel centers' properties, we can reconstruct the intensity and location of each pixel as follows:

$$\hat{a}_{i,j} = \frac{\sum_{m=1}^M \left(\hat{a}_m \cdot q_{i,j}^m\right)}{\sum_{m=1}^M q_{i,j}^m}, \quad \hat{p}_{i,j} = \frac{\sum_{m=1}^M \left(\hat{p}_m \cdot q_{i,j}^m\right)}{\sum_{m=1}^M q_{i,j}^m} \tag{3}$$

where $\hat{a}_{i,j}$ and $\hat{p}_{i,j}$ are, respectively, the reconstructed intensity and position of pixel $f_{i,j}$. $\sum_{m=1}^M q_{i,j}^m$ is the normalization constant, meaning that the total probability that pixel $f_{i,j}$ belongs to all superpixels is one. The physical meaning of (3) is that the reconstructed feature of each pixel is the weighted average of the features of all superpixels related to it. Like (2), we can also write the above equation in the form of matrix multiplication

$$\hat{\Phi} = \widehat{\mathbf{Q}} \begin{bmatrix} \hat{a}_1 & \hat{p}_1 \\ \vdots & \vdots \\ \hat{a}_M & \hat{p}_M \end{bmatrix} = \widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^T \Phi \tag{4}$$

where $\widehat{\mathbf{Q}}$ is the normalized association matrix $\mathbf{Q}$ satisfying $\sum_{m=1}^M q_{i,j}^m = 1$. Here, $\hat{\Phi} \propto \mathbb{R}^{H\times W\times 3}$ is named as reconstructed intensity and position matrix. Its element at ith row, jth column represents the reconstructed intensity and location vector of pixel $f_{i,j}$.

The ideal association map can merge the neighbor and similar pixels into the same superpixels. In this case, the initial property matrix Φ will be very close to reconstructed property matrix $\hat{\Phi}$. From this, the loss can be defined as the difference between original and reconstructed properties of all pixels

$$L_{\mathrm{recons}}^{\mathrm{u}} = \mathcal{L}\left(\hat{\Phi}, \Phi\right) = \left\| \widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^T \Phi - \Phi \right\|_2 \tag{5}$$

where $L_{\mathrm{recons}}^{\mathrm{u}}$ is named as reconstruction loss. $\|\cdot\|_2$ denotes L2-norm. Reconstruction loss encourages the trained superpixel prediction network to group pixels with similar property as far as possible.

Note that reconstruction loss $L_{\mathrm{recons}}^{\mathrm{u}}$ does not require the landcover labels of SAR images. For the images known the pixel-wise ground truth, we can replace the intensity and coordinates values of pixels in (5) with their one-hot encoding vector of landcover labels, expressed as $\Phi_L \propto \mathbb{R}^{H\times W\times R}$. Here, R represents the number of landcovers. The (i, j)th element of Φ_L is a R-dimensional vector, which denotes the landcover label (as one-hot encoding) of pixel $f_{i,j}$. The reconstruction loss from labeled images can be written as

$$L_{\mathrm{recons}}^{\mathrm{label}} = \mathcal{L}\left(\hat{\Phi}_L, \Phi_L\right) = E\left(\widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^T \Phi_L, \Phi_L\right) \tag{6}$$

where cross-entropy E(,) is chosen as the distance measure. $L_{\mathrm{recons}}^{\mathrm{label}}$ is named as one-hot encoding cross-entropy loss function.

Similar with SLIC, we also introduce the compactness loss to encourage the superpixels to be spatially compact, which is defined as

$$L_{\mathrm{compact}} = \frac{\sum_{m=1}^M \sum_{(i,j)\in \mathrm{I}} \left\| \hat{p}_m - p_{i,j} \right\|_2}{H\cdot W\cdot M} \tag{7}$$

where $\|\cdot\|_2$ is L2-norm. The compactness loss can minimize the mean spatial distance between the position vector of superpixel m and all pixels inside it. Hence, the superpixel will have lower spatial variance.

Finally, the superpixel prediction loss function has three terms

$$L_{\mathrm{sp}} = L_{\mathrm{recons}}^{\mathrm{u}} + \alpha \cdot L_{\mathrm{recons}}^{\mathrm{label}} + \beta \cdot L_{\mathrm{compact}} \tag{8}$$

where α, β are weights balancing the three terms. The first two terms encourage the trained model to group pixels with similar property, and the third term enforces the superpixels to be spatially compact.

III. Dissimilarity Measurement of Adjacent Superpixels

The superpixel generation network described above has over-segmented an image into soft superpixels. We express the superpixels from the same image as an undirected weighted graph model $G(V, E)$. The vertices V and edges E represent superpixels and their position relation, respectively. The neighboring superpixels are connected by weighted edges in the graph and each vertex is self-connected. Let $\mathbf{S} \in \mathbb{R}^{M\times M}$ and $\mathcal{D} \in \mathbb{R}^{M\times M}$ represent the self-connected affinity matrix and degree matrix of $G(V, E)$. The weights between v_α and v_β (i.e., $s_{\alpha\beta}$) is defined as

$$s_{\alpha\beta} = \exp\left\{-\sigma_0\left(v_\alpha, v_\beta\right)\right\} \tag{9}$$

where $\sigma_0(\ )$ denotes the dissimilarity measure of two superpixels. If v_α and v_β are not connected, $s_{\alpha\beta}$ will be zero, otherwise $s_{\alpha\beta} \in (0, 1]$. In particular, we define the weight of self-connected edge is one, i.e., $s_{\alpha\alpha} = 1$. The larger $s_{\alpha\beta}$ is, the more similar the statistical characteristics of the two superpixels are, and the more likely the two superpixels belongs to the same landcovers. Obviously, superpixels with a smaller dissimilarity should be merged prior to those with a larger one.

The similarity measure of adjacent superpixels has a great influence on the quality of final segmentation results. Yu *et al.* [37] took the brightness and texture feature distance

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of two superpixels as the dissimilarity. Zhang *et al.* [38] used the Kuiper's distance between two regions' histograms to measure the difference. However, the computations are not efficient. In this article, we present a differentiable, exchangeable, and computationally simple regional backscattering dissimilarity measure.

A superpixel is a group of homogeneous pixels, so the dissimilarity measurement needs to first consider the difference in the statistical intensity/amplitude of the two superpixels. In SAR images, superpixels with similar backscatter will have close statistical intensity or amplitude values. Assume that two superpixels v_α and v_β are homogeneous and have constant backscattering m_α and m_β. Generally, m_α and m_β can be estimated by their respective sample means. The sample means of two superpixels can be obtained by

$$\bar{v}_\alpha = \frac{\sum_{(i,j)\in \mathrm{I}} a_{i,j} \cdot q^m_{i,j}}{\sum_{(i,j)\in \mathrm{I}} q^m_{i,j}}, \quad \bar{v}_\beta = \frac{\sum_{(i,j)\in \mathrm{I}} a_{i,j} \cdot q^\beta_{i,j}}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}}. \tag{10}$$

Hoverer, due to the multiplicative speckle noise within SAR images, there are certain differences between sample mean $\bar{v}_\alpha$ and real backscattering m_α.

Based on the multiplicative speckle model of SAR images [25], the gaps between $\bar{v}_\alpha, \bar{v}_\beta$ and m_α, m_β can be written by

$$\bar{v}_\alpha - m_\alpha = \frac{m_\alpha}{\sum_{(i,j)\in \mathrm{I}} q^\alpha_{i,j}} \sum_{(i,j)\in \mathrm{I}} \left(\varepsilon_{i,j} - 1\right) \tag{11}$$

$$\bar{v}_\beta - m_\beta = \frac{m_\beta}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}} \sum_{(i,j)\in \mathrm{I}} \left(\varepsilon_{i,j} - 1\right) \tag{12}$$

where $\varepsilon_{i,j}$ represents the speckle in pixel $f_{i,j}$. The mean and variance of $\varepsilon_{i,j}$ are one and $(4-\pi)/(L\pi)$ in the L-look SAR amplitude image [39]. As speckle is independent and identically distributed [40], the variances of the two estimates can be written as

$$\mathrm{var}\{(\bar{v}_\alpha - m_\alpha)\} = \frac{(4-\pi)}{L\pi} \frac{m_\alpha^2}{\sum_{(i,j)\in \mathrm{I}} q^\alpha_{i,j}} \tag{13}$$

$$\mathrm{var}\left\{\left(\bar{v}_\beta - m_\beta\right)\right\} = \frac{(4-\pi)}{L\pi} \frac{m_\beta^2}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}}. \tag{14}$$

Actually, random variable $\bar{v}_\alpha - \bar{v}_\beta$ is an estimate of the difference $m_\alpha - m_\beta$. The variance of $\bar{v}_\alpha - \bar{v}_\beta$ can be derived from (13) and (14)

$$\mathrm{var}\left\{\left(\bar{v}_\alpha - \bar{v}_\beta\right) - \left(m_\alpha - m_\beta\right)\right\} = \frac{(4-\pi)}{L\pi}\left(\frac{m_\alpha^2}{\sum_{(i,j)\in \mathrm{I}} q^\alpha_{i,j}} + \frac{m_\beta^2}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}}\right). \tag{15}$$

The backscattering dissimilarity between two superpixels $\sigma_0(v_\alpha, v_\beta)$ can be defined as the normalized $\bar{v}_\alpha - \bar{v}_\beta$ by its standard deviation [25]

$$\sigma_0\left(v_\alpha, v_\beta\right) = \frac{\sqrt{L\pi}\,|\bar{v}_\alpha - \bar{v}_\beta|}{\sqrt{4-\pi}\sqrt{\frac{\bar{v}_\alpha^2}{\sum_{(i,j)\in \mathrm{I}} q^\alpha_{i,j}} + \frac{\bar{v}_\beta^2}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}}}}. \tag{16}$$

Note that here m_α and m_β in (15) are replaced by their estimates $\bar{v}_\alpha$ and $\bar{v}_\beta$. Removing the absolute value calculating $|\bar{v}_\alpha - \bar{v}_\beta|$, $\sigma_0(v_\alpha, v_\beta)$ can be rewritten as

$$\sigma_0(v_\alpha, v_\beta) = \frac{\sqrt{L\pi}\left(1 - \min\left\{\bar{v}_\alpha/\bar{v}_\beta, \bar{v}_\beta/\bar{v}_\alpha\right\}\right)}{\sqrt{4-\pi}\sqrt{\frac{\left(\min\left\{\bar{v}_\alpha/\bar{v}_\beta, 1\right\}\right)^2}{\sum_{(i,j)\in \mathrm{I}} q^\alpha_{i,j}} + \frac{\left(\min\left\{\bar{v}_\beta/\bar{v}_\alpha, 1\right\}\right)^2}{\sum_{(i,j)\in \mathrm{I}} q^\beta_{i,j}}}} \tag{17}$$

$\bar{v}_\alpha$ is essentially the same as $\hat{a}_m$ in (1), which are both the weighted average of all pixels related to the superpixel. It can be seen that $\sigma_0(v_\alpha, v_\beta)$ is a function of $\bar{v}_\alpha/\bar{v}_\beta$, and hence it is robust to multiplicative speckle noise. Furthermore, it is differentiable and exchangeable and can be simply computed.

IV. Fast Superpixel Merging Based on K-Order Connectivity

In the traditional superpixel prediction algorithms, each pixel is uniquely assigned to a certain superpixel. But for the superpixel generation algorithm used in this article, pixels are allocated to multiple superpixels with different probabilities at the same time. This requires us to improve the subsequent superpixel feature extraction and merging algorithms. First, for a certain superpixel, all the pixels that may belong to it must be considered; second, the algorithm must be differentiable, so that gradient descent will be able to make a progress.

In this section, we propose a computationally simple and differentiable superpixel merging algorithm to segment the SAR images. This method develops from the concept of the number of walks in graph theory.

A. K-Order Connectivity in Graph Theory

The concept of the number of walks in graph theory [41] gives an algebraic method to calculate exactly how many paths of a given length exist between two vertices in a graph using the adjacency matrix. Let $\mathbf{A}$ denote the adjacency matrix for this graph and $\mathbf{A}^{(k)}$ the k_{th} power of $\mathbf{A}$. The elements of $\mathbf{A}$ are one or zero and the elements of $\mathbf{A}^{(k)}$ are also nonnegative integers. According to the concept of number of walks, the entry $a^{(k)}_{\alpha\beta}$ of $\mathbf{A}^{(k)}$ represents the total number of paths of length k from v_α to v_β in the graph. For example, there is a self-connected unweighted graph containing four vertexes $\{P_1, P_2, P_3, P_4\}$ shown in Fig. 3(a). In order to find the number of paths of length two from P_3 to P_3, we need to calculate the entry $a^{(2)}_{33}$ of $\mathbf{A}^{(2)}$. Looking at the $\mathbf{A}^{(2)}$ given in Fig. 3(b), the entry $a^{(2)}_{33}$ is three, meaning that there are exactly three paths of length two from P_3 to P_3. These paths are $P_3 \to P_2 \to P_3$, $P_3 \to P_4 \to P_3$, and $P_3 \to P_3 \to P_3$.

Fig. 4(a) shows a weighted undirected graph with five vertexes and its self-connected affinity matrix $\mathbf{S}$. As we mentioned in dissimilarity measurement section, $s_{\alpha\beta}$ ranges from 0 to 1. The larger the $s_{\alpha\beta}$ is, the more similar the statistical characteristics of the two superpixels are. $s_{\alpha\beta} = 1$ indicates that two superpixels v_α and v_β are exactly the same. As shown in Fig. 4(a), greater $s_{\alpha\beta}$ means shorter edge in graph.

Yu *et al.* [37] extended the concept of the number of walks for the unweighted graph to that for the weighted self-connected graph. They defined the entry $s^{(k)}_{\alpha\beta}$ of $\mathbf{S}^{(k)}$ as k-order connectivity between superpixel v_α and v_β. Particularly, the diagonal entry $s^{(k)}_{\alpha\alpha}$ is defined as k-order connectivity of vertex v_α. Note that at this time the value of $s^{(k)}_{\alpha\beta}$ becomes a real number, not a nonnegative integer.

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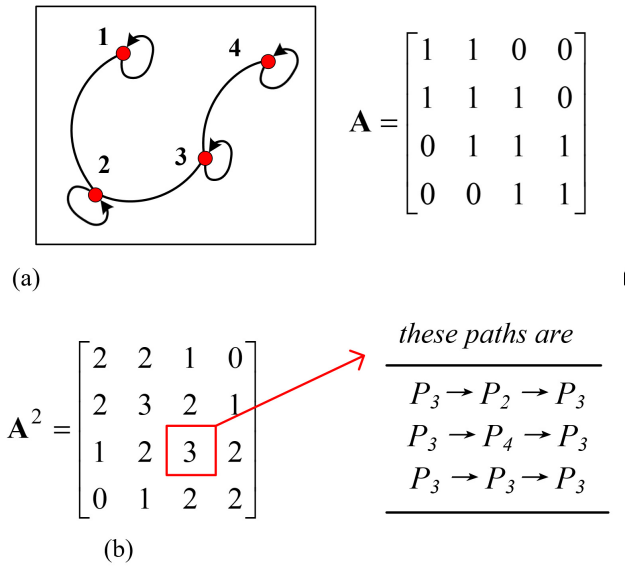


Fig. 3. Simple example to illustrate the number of walks in graph: (a) self-connected undirected graph with four vertexes and (b) entry $a_{33}^{(2)}$ of $\mathbf{A}^{(2)}$ represents the number of paths of length two from P_3 to P_3.

B. Soft Cluster Center Based on K-Order Connectivity

A graph can be regarded as a railway network where each vertex is a station. The k-order connectivity represents how busy the station is. The larger the $s_{\alpha\beta}^{(k)}$, the larger the number of walks of length k from v_α and v_β. It means that there are more stations and railways between v_α and v_β.

For vertex v_α, greater $s_{\alpha\alpha}^{(k)}$ indicates that more stations are connected to station v_α. If $s_{\alpha\alpha}^{(k)}$ is greater than $s_{\alpha\beta}^{(k)}(\beta = 1, \ldots, M, \beta \neq \alpha)$, v_α a will be the busiest in the nearby stations. Obviously, v_α will be more likely to be a transportation hub or a cluster center of data. Motivated by this fact, Geng and Tang [42] gave the definition of k-order hard cluster center.

Definition 1 (k-Order Hard Cluster Center): Vertex v_α will be defined as a k-order hard cluster center of the graph if $s_{\alpha\alpha}^{(k)}$ is greater than $s_{\alpha\beta}^{(k)}(\beta = 1, \ldots, M, \beta \neq \alpha)$, denoted as

$$v_\alpha^c = \begin{cases} 1, & \text{if } s_{\alpha\alpha}^{(k)} > s_{\alpha\beta}^{(k)}(\beta = 1, \ldots, M, \beta \neq \alpha) \\ 0, & \text{else} \end{cases} \tag{18}$$

where $v_\alpha^c \in \{0, 1\}$ is an indicator of superpixel v_α and $v_\alpha^c = 1$ means that v_α is a clustering center. Otherwise, v_α is a noncluser-center. Here, definition include word "hard" because v_α^c can only be zero or one.

Fig. 4(b) shows the value of the third power matrix $\mathbf{S}^{(3)}$. For intuitive perception and easy understanding, here we omit the decimal part of the elements $s_{\alpha\beta}^{(3)}$ and draw a colormap to indicate the elements' values. Since v_1 has small similarity with v_4, this graph can be intuitively divided into two subgraphs, namely, $\{v_1, v_2, v_3\}$ and $\{v_4, v_5\}$. According to the given $\mathbf{S}^{(3)}$ and definition of hard cluster centers, v_2 and v_4 are selected as the k-order hard cluster centers of the graph, which is consistent with manual segmentation results.

However, note that (18) is essentially a Heaviside step function, which cannot be used as an activation function in deep neural networks. One reason is that the backpropagation algorithm requires activation functions that are differentiable. However, the step function is nondifferentiable when $x = 0$ and its derivative is zero elsewhere. The gradient descent will not be able to make a progress in updating the weights. Meanwhile, the purpose of the training the neural network is to learn the values of the weights and biases so that the network could make correct predictions. In order to continuously tweak the values of weights and bias toward capturing the optimal estimation, we hope that small changes in weight or bias will cause corresponding changes in output. The step function clearly cannot meet such requirements.

A main contribution of our approach is to convert the cluster center selecting operation into a differentiable one. Instead of computing hard cluster center indicator, we propose to compute the soft-indicator.

Definition 2 (k-Order Soft Cluster Center): If $s_{\alpha\alpha}^{(k)}$ is much greater than all $s_{\alpha\beta}^{(k)}(\beta = 1, \ldots, M, \beta \neq \alpha)$, this superpixel is more likely to be a cluster center. Let soft-indicator v_α^c represent the probability that a superpixel is selected as a cluster center, which is calculated by

$$v_\alpha^c = \text{Sigmoid}\left\{\min\left[\left(s_{\alpha\alpha}^{(k)} - s_{\alpha 1}^{(k)}\right), \left(s_{\alpha\alpha}^{(k)} - s_{\alpha 2}^{(k)}\right), \ldots, \left(s_{\alpha\beta}^{(k)} - s_{\alpha M}^{(k)}\right)\right]\right\} \tag{19}$$

where Sigmoid indicates a logistic sigmoid function.

The value of new v_α^c ranges from zero to one. If $s_{\alpha\alpha}^{(k)}$ is much greater than all $s_{\alpha\beta}^{(k)}(\beta = 1, \ldots, M, \beta \neq \alpha)$, the value of v_α^c will be close to one and this superpixel is more likely to be a cluster center. The indicators of all superpixels from the same graph can form a vector $\mathbf{V}^c = [v_1^c, v_2^c, \ldots, v_M^c] \in \mathbb{R}^{M\times 1}$, named soft cluster centers indicator vector. In this case, the cluster center finding becomes differentiable and can be implemented using deep networks.

The merging process is based on the soft cluster center for each $\mathbf{S}^{(k)}$, and after selecting the soft cluster centers according to (19), we next assign the remaining vertexes to the corresponding cluster centers based on the connectivity of vertices. Intuitively, a vertex should be assigned to the cluster center with which it has the greatest connectivity. As shown in Fig. 4(c), for v_1, the third-order connectivity of between v_1 and v_2 is greater than that between v_1 and v_4. Hence, vertex v_1 is assigned to v_2. Similarly, for vertex v_3, its third-order connectivity with v_2 is around three, while it is not connected with v_4 so that v_3 finally is assigned to cluster center v_2.

In order to implement this merging process using the deep convolution network, for any noncluster-center v_α, we introduce a vector $\mathbf{v}_\alpha^* = [v_{\alpha 1}^*, \ldots, v_{\alpha M}^*] \in \mathbb{R}^{M\times 1}$ to represent the probability that v_α is assigned to all cluster centers. $\mathbf{v}_\alpha^*$ can be calculated as follows:

$$\begin{aligned} \mathbf{v}_\alpha^* &= \text{softmax}\left\{\mathbf{S}_\alpha^{(k)} \odot \mathbf{V}^c\right\} \\ &= \text{softmax}\left\{\left[s_{\alpha 1}^{(k)}, \ldots, s_{\alpha M}^{(k)}\right] \odot \left[v_1^c, \ldots, v_M^c\right]\right\} \\ &= \text{softmax}\left\{\left[s_{\alpha 1}^{(k)} \cdot v_1^c, \ldots, s_{\alpha M}^{(k)} \cdot v_M^c\right]\right\} \end{aligned} \tag{20}$$

where $\mathbf{S}_\alpha^{(k)}$ is the α_{th} row of matrix $\mathbf{S}^{(k)}$ and $\odot$ denotes the element-wise product. $\mathbf{V}^c$ has been obtained in (19). Obviously, the sum of all elements of $\mathbf{v}_\alpha^*$ is one and its elements

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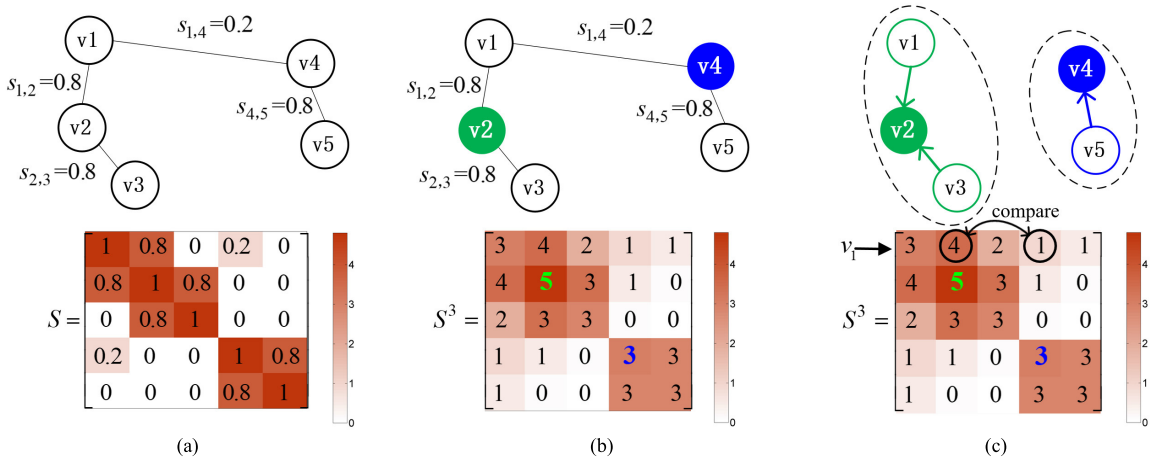


Fig. 4. Example illustrating the differentiable superpixel merging method based on the k-connectivity of vertexes: (a) weighted self-connected undirected graph with five vertexes and its affinity matrix $\mathbf{S}$. Superpixels with higher similarity appear to be closer in space; (b) soft cluster centers of the graph based on Definition 2, i.e., v_2 and v_4; and (c) merging process based on soft cluster center. The remaining vertexes are assigned to the cluster centers with which it has the greatest k-order connectivity. v_1 and v_3 are assigned to v_2, while v_5 is merged with v_4.

$s_{\alpha\beta}^{(k)} \cdot v_\beta^c$ indicate the probability that a superpixel v_α assigned to superpixel v_β. Only when v_β has a high probability of being a cluster center (that is the value of v_β^c close to 1) and v_α has a larger k-order connectivity with v_β, $s_{\alpha\beta}^{(k)} \cdot v_\beta^c$ will have a larger value. In this case, v_α is more likely to be merged with the cluster center v_β. Finally, assuming that $v_{\alpha\gamma}^*$ is the largest element of $\mathbf{v}_\alpha^* = [v_{\alpha 1}^*, \ldots, v_{\alpha M}^*]$, v_α will be merged with v_γ.

According to (20), the segmentation result of our clustering method is determined by the affinity matrix $\mathbf{S}$ and number of walks, i.e., k. Essentially k is a scale adjustment factor, which determines how far apart vertexes will affect each other in the clustering process. Fig. 5 shows the number of cluster centers after merging on five real SAR images when k takes different values. The images' sizes are 1024 × 1024 and they are initially over-segmented into 1000 superpixels. As can be seen in Fig. 5, when k first gradually increases from zero, the number of cluster centers will drop rapidly. When k is greater than a certain value, the number of cluster centers will remain stable. The experimental results show that when k is set as between 100 and 200, the proposed method can maintain satisfactory segmentation results, proving our method has good robustness to the values of k. This is because our superpixel generation network can adaptively adjust the over-segmentation results, Even if k is not given the optimal initial value, the superpixel generation network can adjust the shape of the superpixels during the training phase until capturing the optimal affinity matrix under the current value of k.

Fig. 6 shows our merging results on an SAR image under different number of walks. The SAR image in Fig. 6(a) has two landcover classes: water and land. The initial number of cluster centers is equivalent to the number of superpixels. When k is small, $s_{\alpha\alpha}^{(k)}$ mainly reflects the connectivity of the v_α with spatially nearby vertexes. The superpixels are mainly affected by their neighboring superpixels. Vertex v_α with greater $s_{\alpha\alpha}^{(k)}$ value can be considered as the local cluster centers. Fig. 6(b) indicates the merging results when $k = 10$. At this time, 118 vertexes are selected as the cluster centers. Fig. 6(c) and (d) illustrate the merging results when $k = 20$ and 30.

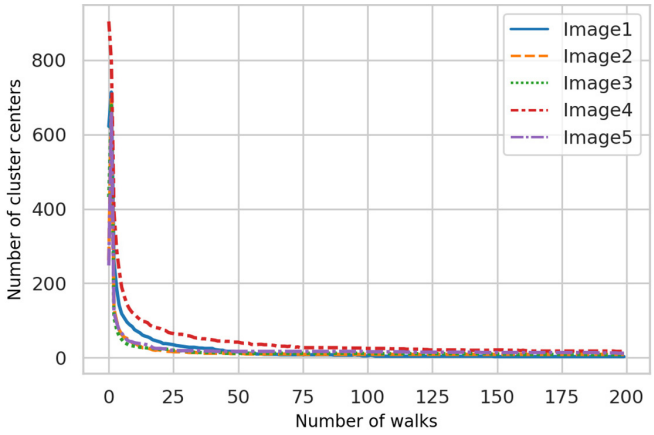


Fig. 5. Number of cluster centers after merging with different number of walks. When k is greater than a certain value, the number of cluster centers will remain stable.

As the value of k increases, that is, the length of the walks continues to increase, some vertexes that are farther away will be considered in the walk. At this time, the vertexes with the larger $s_{\alpha\alpha}^{(k)}$ become the global cluster center and the number of cluster centers will gradually decrease. In Fig. 6(d), there are only six global cluster centers left. Water and land regions are correctly segmented.

In order to make the merging results as close as possible to the ground truth, we design a superpixel merging loss. First, for the images in the training set, we need to convert their pixel-level landcover labels into superpixel-wise labels. Here, we apply the majority voting strategy, that is, the category of a superpixel is the most frequent class in this superpixel.

Assuming that the label of superpixel v_α is L_α, after getting $\mathbf{v}_\alpha^*$ in (20), we can predict the label of the cluster center v_β by

$$\tilde{L}_\beta = \sum_{\alpha=1}^{M} \frac{v_{\alpha\beta}^* \cdot L_\alpha}{\sum_{\alpha=1}^{M} v_{\alpha\beta}^*} \tag{21}$$

where $\tilde{L}_\beta$ denotes the predicted label of v_β. It is the weighted average of the categories of all nodes in this cluster. Based

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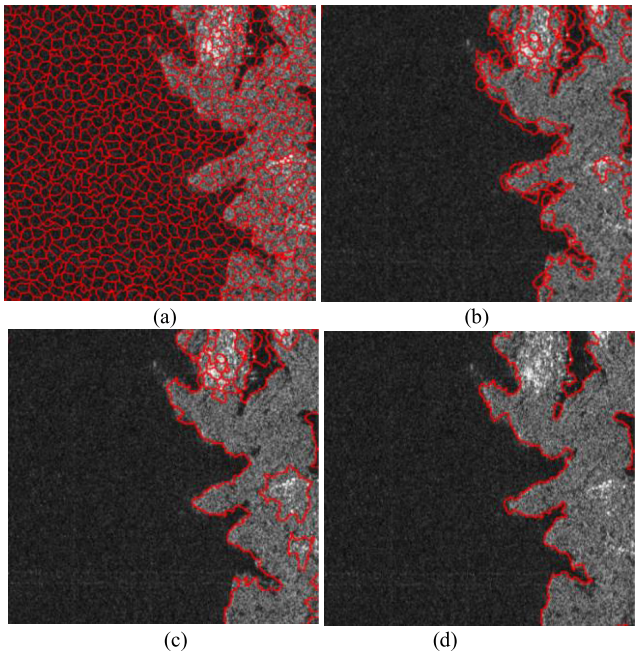


Fig. 6. Clustering process under different number of walks: (a) superpixel generation using deep network. The initial number of cluster centers is equivalent to the number of superpixels and (b)–(d) merging results of our method based on 10-, 20-, and 30-order connectivity, respectively. After merging, the number of cluster centers is 118, 41, and 6.

on this, we can further reconstruct the categories of all noncluster-center nodes as follows:

$$\hat{L}_\alpha = \sum_{\beta=1}^{M} \frac{v^*_{\alpha\beta} \cdot \tilde{L}_\beta}{\sum_{\beta=1}^{M} v^*_{\alpha\beta}} \tag{22}$$

where $\hat{L}_\alpha$ essentially represents the weighted average of all cluster centers' labels. If the clustering results represented by $\mathbf{v}^*_\alpha$ are correct, that is, the vertexes with the same categories are merged, $\hat{L}_\alpha$ will be very close to the real label L_α.

Then, we can calculate the corresponding superpixel merging loss function

$$L_{\text{merge}} = \sum_{\alpha=1}^{M} \mathcal{L}\left(\hat{L}_\alpha, L_\alpha\right) \tag{23}$$

where $\mathcal{L}$ is the cross-entropy function. In summary, the loss function of this end-to-end segmentation method can be written as

$$\begin{aligned} L &= L_{\text{sp}} + L_{\text{merge}} \\ &= L^{\text{u}}_{\text{recons}} + \alpha \cdot L^{\text{label}}_{\text{recons}} + \beta \cdot L_{\text{compact}} + \lambda \cdot L_{\text{merge}} \\ &= \left\| \widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^T \Phi - \Phi \right\|_2 + \alpha \cdot E\left(\widehat{\mathbf{Q}}\tilde{\mathbf{Q}}^T \Phi_L, \Phi_L\right) \\ &\quad + \beta \cdot \frac{\sum_{m=1}^{M} \sum_{(i,j)\in \mathrm{I}} \left\| \hat{p}_m - p_{i,j} \right\|_2}{H \cdot W \cdot M} + \lambda \cdot \sum_{\alpha=1}^{M} \mathcal{L}\left(\mathbf{v}^*_\alpha, L^*_\alpha\right) \end{aligned} \tag{24}$$

where λ is a weight to balance the merging loss term. It can be seen from (24) that the four loss terms guide the deep network to adjust the superpixels' shape according to the boundary and landcover labels information in the SAR image. Table I shows the training step of our proposed superpixel-wise segmentation algorithm.

TABLE I
TRAINING STEPS OF PROPOSED SEGMENTATION METHOD

Input: Unlabeled and labeled SAR images $\mathbf{I} \propto \mathbb{R}^{H\times W}, \mathbf{I}_L \propto \mathbb{R}^{H\times W}$
Output: Superpixel-wise segmentation results

for Number of training iterations **do**

1. Calculate unlabeled/labeled intensity-coordinate matrix Φ, Φ_L.
2. Superpixel generation network takes Φ, Φ_L as inputs and outputs soft association maps $\mathbf{Q}$.
3. Model the over-segmented SAR image into a self-connected undirected graph $G(V, E)$.
4. Calculate weighted self-connected affinity matrix $\mathbf{S}$ of each image as (9) and (17).
5. Define the K-order connectivity matrix between connected vertexes, i.e., $\mathbf{S}^{(k)}$.
6. Calculate the soft cluster center indicator vector $\mathbf{v}^c$ to determine cluster centers based on $\mathbf{S}^{(k)}$ as (19).
7. Merge remaining superpixel to their cluster centers guided by $\mathbf{S}^{(k)}$ and $\mathbf{v}^c$ as (19).
8. Calculate superpixel generation loss L_{sp} as (8) and superpixel merging loss L_{merge} as (24).
9. Update the parameters of superpixel generation networks based on L_{sp} and L_{merge}.

End for

C. Discussion on the Computational Complexity

Considering that this article mainly pays attention to the superpixel merging, here we only analyze the computational complexity of the merging stage, whose computation cost is composed of the calculation of dissimilarity measures and superpixel merging. In the dissimilarity measure calculation, the time complexity can be denoted as $O(K \cdot M)$, where K represents the number of adjacent superpixels for each superpixel and M is the number of superpixels in an input SAR image. Generally, for a superpixel, its unnecessary to compute the dissimilarity measures with all other superpixels. We constraint the number of adjacent superpixels adopting the strategy in [33] and [36]. The input image is first partitioned using a regular grid and each grid cell is considered as an initial superpixel. Only the cluster centers located in surrounding eight grid cells are regarded as its adjacent superpixel. Consequently, the K can be simplified to eight and time complexity of constructing the similarity matrix can be rewritten as $O(M)$.

The execution of soft clustering mainly consists of three parts: calculating kth power of the similarity matrix, determining cluster centers, and merging superpixel to their cluster centers. Compared to the first item, the computational complexity of the latter two items can be negligible. The computational complexity of the kth power of an $M \times M$ matrix is $O(k \cdot M^3)$. The overall time complexity of the proposed merging algorithm can be denoted as $O(k \cdot M^3)$, which is quite related to the superpixel number and number of walks.

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It seems like that the time complexity of our proposed merging approach will increase dramatically with the increasing of superpixel number. However, the matrix multiply can occur in parallel with multiple streaming processors of graphics processing unit (GPU). It means that we can apply parallel computing in the soft clustering stage to reduce the time of computing.

V. Experimental Results and Analysis

This section first describes the used three SAR datasets and implementation details of our method. Then, we conduct the segmentation experiments and quantitatively compare with state-of-the-art methods. Finally, we give the running time of existing methods on the same datasets to quantitatively evaluate the computational complexity of the proposed algorithm.

A. Dataset

1) *Fangchenggang Dataset*: This dataset was acquired by airborne SAR sensors (Ku-band, HV polarization) at Fangchenggang Town (N21°41′ 22.66″, E108°21′ 2.64″), Guangxi Province, China, in March 2016. It contains 57 SAR images with size of 1024 × 1024 pixels and resolution of 2 m. Training set has six and testing set consists of 51 images. This dataset has four landcover classes: farmland, water, urban and background.
2) *Gaofen Dataset*: Gaofen dataset is released in 2020 Challenge on Automated High-Resolution Earth Observation Image Interpretation [43]. It totally contains 500 high-resolution spaceborne SAR images with L-band, FSII mode and HV single polarization. The training set has 100 images and the remaining are testing set. The size of each image is 512 × 512, and the resolution is from 10 to 30 m. The ground truth maps contain six land-cover types: water-body, buildings, industrial area, lawn, bare land, and background.
3) *PoyangLake Dataset*: The image data used for this study is acquired by spaceborne GF-3 sensors operating in the L-band, FSII mode and HH single polarization. As shown in Fig. 7, the imaging range is located at Poyang Lake (E116.2, N29.2), Jiangxi Province, China, on July 8 and 13, 2020. The dataset consists of 1511 SAR images with a size of 1024 × 1024 and a spatial resolution of 10 m × 10 m. Among them, 190 images belong to the training set and the remaining is testing set. This dataset is labeled manually according to the Google Earth maps, including two landcover classes: water and land.

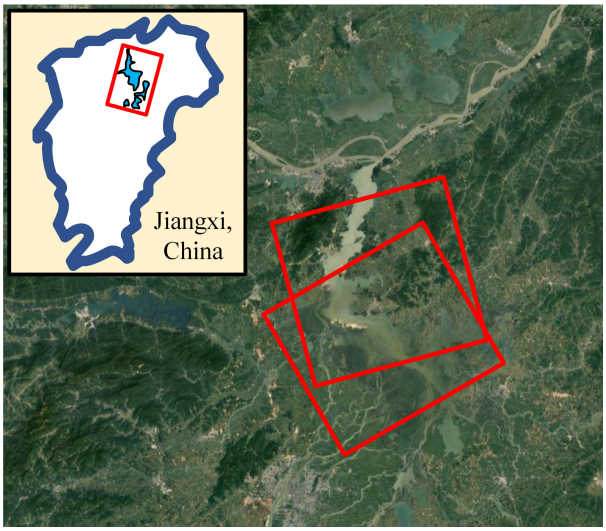


Fig. 7. Imageable areas of the PoyangLake dataset.

B. Experimental Settings

1) *Implementation Details*: Our superpixel generation network uses an encoder–decoder design with skip connections, which is the same with [33], consisting of five convolution layers and four deconvolution layers. The five convolution layers contain 16, 32, 64, 128, and 256 convolution kernels with a size of 3 × 3, respectively. We use a grid with cell size 16 × 16 as the initial superpixels. The model is implemented with PyTorch and optimized using Adam with learning rate = 1×10^{-4} and batch-size = 100. According to the suggestions in [33] and [36], we set $\alpha = 0.01$, $\beta = 0.1$, and $\lambda = 0.01$ in all experiments. In the superpixel merge phase, we set the number of walks as 100.
2) *Evaluation Metrics:* We mainly introduce three indicators: per-class accuracy, pixel-wise overall precision (OP) [44], pixel-wise overall accuracy (OA) [45], and Cohen's kappa coefficient (κ) to quantitatively compare the performance of various methods. Specifically, OP and OA are the weighted average of recall and precision, respectively, where the weights indicate the proportion of each class of pixels in the ground truth images. The kappa coefficient is defined as

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{25}$$

where p_o is the relative agreement between the segmentation results and the real labels; p_e is the hypothetical probability of a chance agreement. A kappa coefficient closer to one denotes the better agreement between the results and ground truth.
3) *Comparison Methods*: To show the effectiveness of our proposed method, here we choose five recent state-of-the-art SAR image segmentation methods for performance comparison. Three of them are superpixel-wise methods for SAR images and achieve the final segmentation by interactively merging regions. They are, respectively, hierarchical region-merging method with gradually increasing orientated edge strength penalty (GIOESP) [38], fast region-merging approach with RCBLP [26], and newly region-merging method with SRM [28]. The other two segmentation methods are based on representative pixel-wise methods, i.e., SegNet [46] and Deeplab V3+ [47]. SegNet is an encoder–decoder structure semantic segmentation method. The encoder is composed of 13 convolution layers of VGG16 and the decoder maps the features

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to the segmentation results. The key parameters of the SegNet are all consistent with [46]: patch size = 16 × 16, learning rate = 0.0001, regular regularization parameter = 0.0005, batch size = 300, and training epoch = 100. Deeplab V3+ is a recently proposed semantic segmentation network, which is mainly composed of ResNet101-V2 and the atrous spatial pyramid pooling (ASPP) module. It can resist the limitation of the fixed size of the input image to extract the high-level multiscale information. Similarly, the hyper-parameters of Deeplab V3+ are consistent with article [47]. All these methods are implemented ten times on each dataset, and the mean accuracies and standard deviations over these ten independent implementations are reported.

C. Training Process of Proposed Method

Compared with the existing methods, the main advantage of our method is to continually optimize the superpixel generation and merging parameters through end-to-end training. This part aims at showing the improvement in segmentation performance of our method during the training. Fig. 8 shows the superpixel generation results under various epochs. In order to show this change more intuitively, the input we use is a four-look simulated SAR image. It is generated by adding the multiplicative speckle obeying gamma distribution to an image with different shape texture and carton texture. Fig. 8(a) is one of the simulated images with four-look gamma speckle. These images are generated by adding the multiplicative speckle obeying gamma distribution to an image with different shape texture and carton texture. The image size of 430 × 375. Fig. 8(b)–(d) gives the output of the superpixel generation network when the epoch is 10, 20, and 50, respectively. Some superpixels that falsely adhere the boundaries are marked by blue lines. It can be seen that as the number of epochs increases, the number of false superpixels keeps decreasing and the shapes of the superpixels can capture the boundaries of the SAR image better and better.

Fig. 9 shows the relationship between the number of soft cluster centers and the number of walks under different epochs. When epoch = 10 and 20, the number of cluster centers decreases slowly with the number of walks. This is because the superpixels at this time are not very accurate and many superpixels contain more than two classes of pixels. They are easily considered as new categories in the merging stage and selected as the soft cluster centers. As the epochs of training increase, the accuracy of superpixels continues to improve. Most of the superpixels contain only one type of landcover. At this time, the superpixels can be merged faster. In Fig. 9, when epoch is 30 and 50, the number of cluster centers decreases more rapidly as the number of walks.

D. Segmentation Results

1) Results on Fangchenggang Dataset: The quantitative results obtained by different methods on the Fangchenggang dataset are summarized in Table II, where the highest value in each row is highlighted in bold. We observe that the proposed method achieves the top-level performance among all the methods in terms of OP, OA, and kappa coefficient,

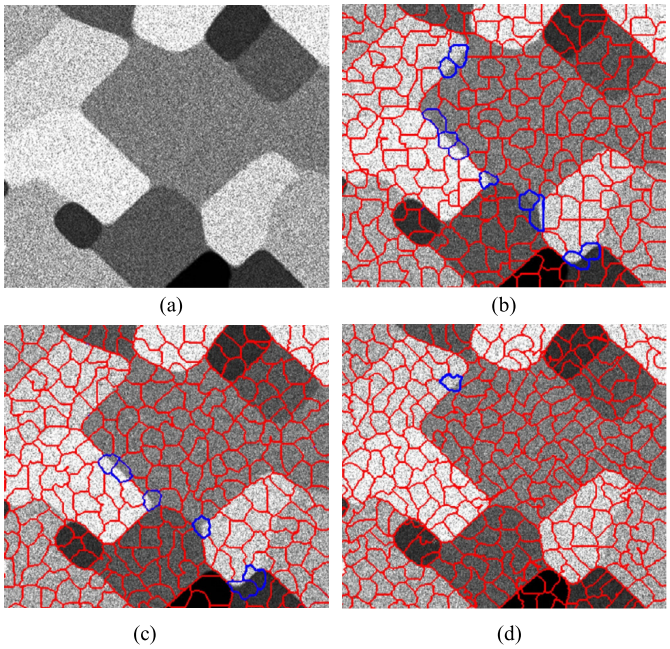


Fig. 8. Change of superpixel generation results under different number of epochs: (a) four-look simulated SAR image and (b)–(d) superpixels from our method when epoch is 10, 20, and 50, respectively. The worry superpixels are marked by blue lines.

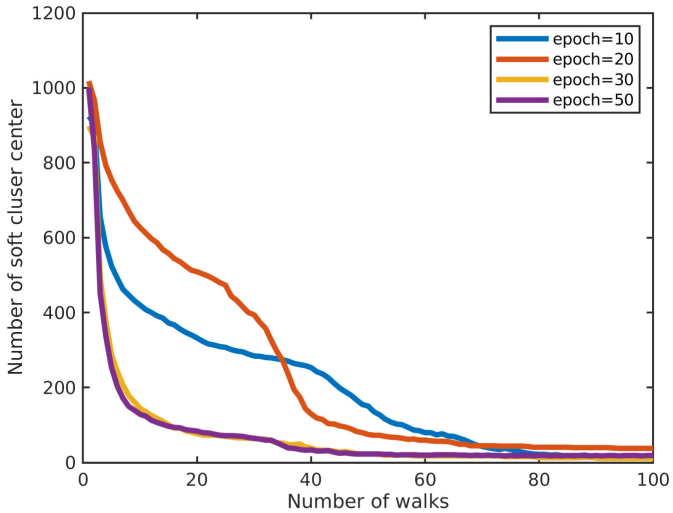


Fig. 9. Relationship between the number of soft cluster centers and the number of walks k under different epochs. When epoch is 30 and 50, superpixels can be merged faster and the merging process is more robust to the value of k.

and the standard deviations are also very small. Therefore, it is reasonable to infer that the proposed method is more stable and effective than the compared methods.

Fig. 10 shows a visual comparison of the segmentation results generated by different methods on the Fangchenggang dataset. Compared with the ground-truth map, due to the lack of spatial context, a lot of isolated falsely segmented pixels occurs in the segmentation maps of by SegNet and Deeplab V3+. Meanwhile, GIOESP, RCBLP, and SRM all take superpixel as processing unit so that their results have better compactness and less false pixels. It should be noted that these superpixel-wise methods require high precision of

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TABLE II

PER-CLASS ACCURACY, OP (%), OA (%), AND KAPPA COEFFICIENT OF THE OVERALL SEGMENTATION ON FANGCHENGGANG DATASET

Class	SegNet	Deeplab V3+	GIOESP	RCBLP	SRM	Ours
Farmland	83.22±0.12	72.85±0.21	86.54±0.26	84.81±0.11	86.56±0.56	**88.35±0.06**
Water	83.22±0.12	76.84±0.22	76.32±0.15	85.17±0.34	87.65±0.43	**91.21±0.14**
Urban	66.32±0.48	76.84±0.17	84.65±0.34	86.24±0.63	86.89±0.78	**89.65±0.13**
Background	86.68±0.17	86.36±0.87	86.89±0.46	87.65±0.29	88.65±0.33	**90.24±0.07**
OP (%)	85.85±0.32	85.58±0.42	86.14±0.35	86.76±0.21	87.68±0.34	**90.10±0.06**
OA (%)	84.90±0.22	82.13±0.26	85.94±0.41	84.69±0.28	86.57±0.44	**89.97±0.11**
Kappa	75.82±0.94	72.43±0.51	77.34±0.36	77.06±0.22	80.84±0.53	**83.24±0.28**

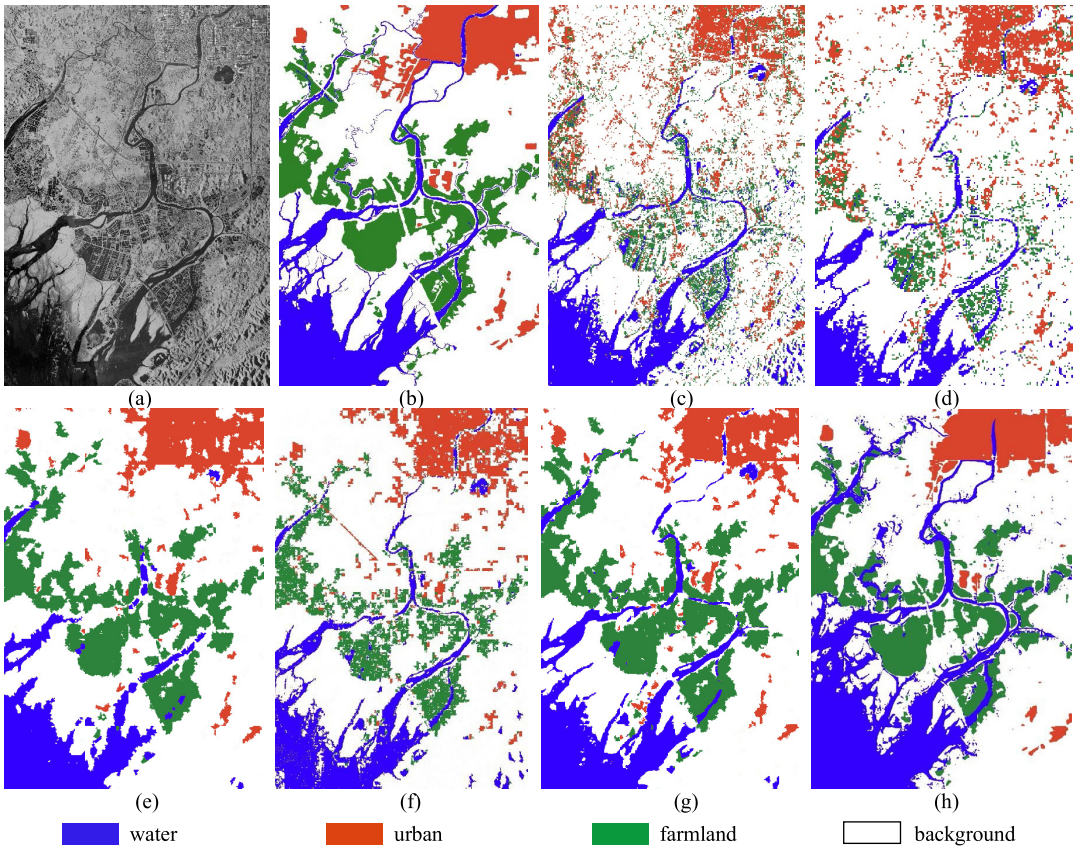


Fig. 10. Segmentation results of the different methods on Fangchenggang data: (a) original SAR images; (b) ground-truth; (c) SegNet; (d) Deeplab V3+; (e) GIOESP; (f) RCBLP; (g) SRM; and (h) proposed method.

superpixels. The accuracy of superpixels determines their segmentation performance to a certain extent. As many pixels in narrow river areas are clustered into the land superpixels, a lot of tributaries are mis-segmented into "background" in their results. These errors are difficult to remove using postprocessing steps such as morphological processing. Comparatively, the result of the proposed method in Fig. 10(h) yields a smoother visual effect and shows fewer mis-segmentations.

The structural similarity index measuring (SSIM) [48] here is also introduced in Fig. 10 to quantify the similarity between the ground truth map in Fig. 10(b) and segmentation images in Fig. 10(c)–(h). SSIM is a multiplicative combination of three terms, namely, the luminance term, the contrast term, and the structural term, which is defined as

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{26}$$

where μ_x, μ_y, σ_x, σ_y, and σ_{xy} are the local means, standard deviations, and cross-covariance for images x, y. c_1 and c_2 are constants set to avoid the denominator being zero, whose values are consistent with that in [48]. The larger the SSIM value, the closer the segmentation results are to ground trues. In view of SSIM, our SSIM value is 0.864, far larger than that of other methods. It is consistent with the above conclusions by visual observing.

2) Results on Gaofen Dataset: Table III illustrates the quantitative evaluation measures of different methods for Gaofen datasets, where the bold fonts denote the best indices. Compared with the Fangchenggang dataset, the SAR images in Gaofen dataset have lower resolution. Meanwhile, here the imaging regions mainly locate at urban areas, so the segmentation task is more challenging. Compared with Table II, the kappa coefficients of individual methods decrease to a certain extent. In Table III, the proposed method still achieves the

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TABLE III
PER-CLASS ACCURACY, OP (%), OA (%), AND KAPPA COEFFICIENT OF THE OVERALL SEGMENTATION ON GAOFEN DATASET

Class	SegNet	Deeplab V3+	GIOESP	RCBLP	SRM	Ours
lawn	76.68±0.84	76.38±0.54	80.26±0.22	79.65±0.35	85.32±0.46	**86.67±0.13**
bare land	81.35±0.16	86.24±0.58	85.35±0.19	**87.16±0.08**	83.51±0.13	86.56±0.83
industrial area	70.36±0.37	79.36±1.64	80.35±0.86	79.64±0.46	**82.48±0.53**	81.51±0.22
water-body	77.26±0.57	81.57±1.11	89.35±0.34	78.58±0.25	84.68±0.25	**84.76±0.34**
buildings	70.91±0.84	80.64±0.74	76.24±0.62	80.36±0.71	87.35±0.45	**88.65±0.17**
background	72.14±0.64	77.68±0.78	82.14±0.18	82.35±0.16	83.57±0.54	**85.16±0.34**
non-image	89.35±0.45	92.31±0.24	87.36±0.64	84.97±0.32	90.54±0.58	**94.35±0.22**
OP (%)	72.46±0.13	78.24±0.34	81.24±0.87	83.65±0.61	84.14±0.44	**87.89±0.21**
OA (%)	72.53±0.33	78.36±0.35	80.54±0.13	81.58±0.40	85.71±0.39	**88.32±0.18**
Kappa	69.20±1.14	73.54±2.36	75.45±0.64	76.98±0.49	77.24±0.48	**80.32±0.38**

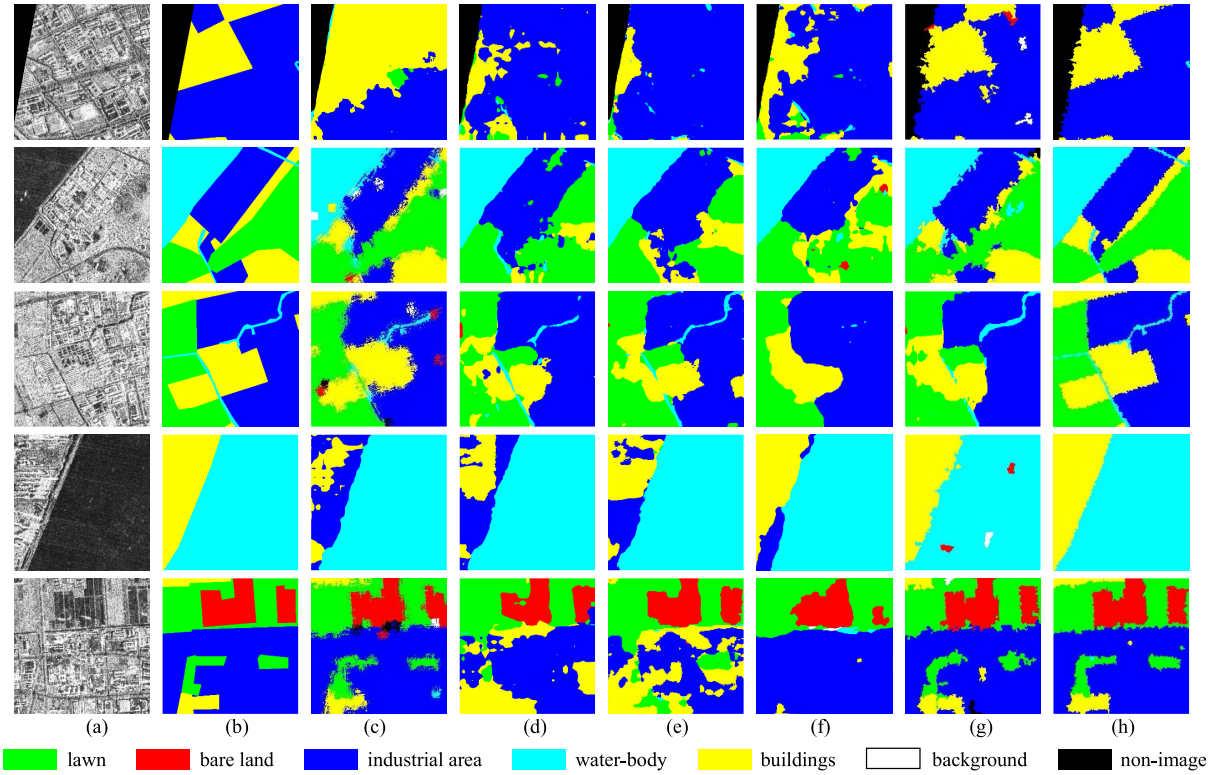


Fig. 11. Segmentation results of the different methods on Gaofen data: (a) original SAR images; (b) ground-truth; (c) SegNet; (d) Deeplab V3+; (e) GIOESP; (f) RCBLP; (g) SRM; and (h) proposed method.

best values in terms of the OP, OA, and kappa coefficient. In can be concluded that the segmentation regions generated by the proposed method are closer to the ground truth than other state-of-the-art methods.

Fig. 11 shows the segmentation results of different methods for a part of images from Gaofen data. From Fig. 11(c) to (d), the pixel-level methods are more susceptible to the speckle noises and cannot well preserve the boundaries within SAR images. DeepLab V3+ has much more parameters than SegNet, so it generally requires more training data. As Gaofen data only has 100 training images, the performance of DeepLab V3+ even is poorer than SegNet. The results of GIOESP, RCBLP, and SRM in Fig. 11(e)–(g) seem better than those of DeepLab V3+ and SegNet. Most of the regions can be correctly discriminated. Nevertheless, there are still some error segmentation results in man-made areas, such as the buildings and industrial targets. The result of the proposed method in Fig. 11(h) is satisfactory, where most of the regions can be correctly discriminated with less false segmentation results. Some small-size buildings can also be well segmented. The SSIM of each method in Fig. 11 is also calculated based on the ground truth map in Fig. 11(b) and corresponding segmentation images in Fig. 11(c)–(h). The proposed method has the highest SSIM (0.853), while the second highest is 0.836. It further proves the superiority of the proposed algorithm.

3) Results on PoyangLake Dataset: Table IV quantitatively compares our algorithm with five state-of-the-art methods on the PoyangLake dataset. It is apparent that the performance of all methods is better than that on the Fangchenggang and Gaofen datasets. The reason is that the PoyangLake dataset has higher spatial resolution and contains less speckle noise than other two datasets and thus is more suitable for segmentation. Note that our method still has better indicator in terms of OA,

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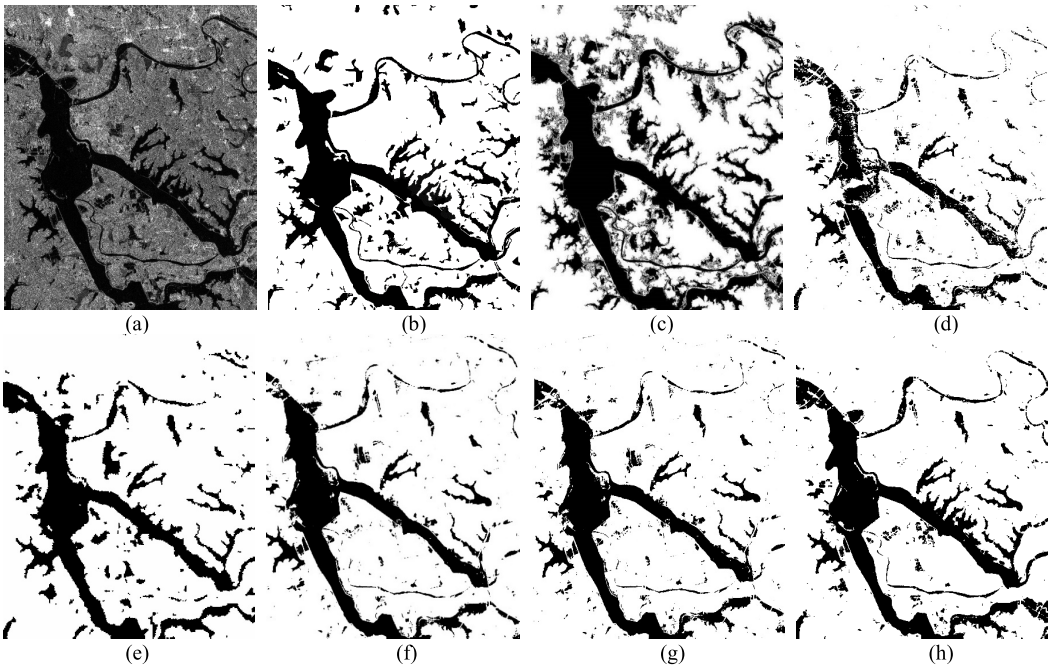


Fig. 12. Segmentation results of the different methods on PoyangLake data: (a) original SAR images; (b) ground-truth; (c) SegNet; (d) Deeplab V3+; (e) GIOESP; (f) RCBLP; (g) SRM; and (h) proposed method.

OP, and kappa coefficient. Only the accuracy for Land regions of our method is slightly less than that of SRM.

Fig. 12(a) is an image of PoyangLake dataset and Fig. 12(b) is the corresponding ground truth. Poyang Lake has complicated water boundaries, greatly increasing the difficulty of segmentation task. In Fig. 12(c) and (d), SegNet and DeepLab V3+ pay more attention to the intensity of the pixels, neglecting the context and boundaries information. For some landcovers with low backscattering coefficients, e.g., airport runways and radar shadow, their pixels in SAR image are easily wrongly predicted as water regions. In Fig. 12(e)–(g), GIOESP, RCBLP, and SRM can detect most of the distinct boundaries in SAR images. The performance of GIOESP, RCBLP, and SRM is certainly restricted by the precision of superpixel generation. Their superpixels cannot preserve some weak boundaries, especially these in the fishpond areas. Hence, there are some over- and under-segmentation areas in their results. In comparison, the proposed method can better preserve the shape of the water areas. Its result contains fewer isolated, overpredicted, and underpredicted pixels. From the perspective of SSIM, our method reaches 0.895, higher than other methods. This is partly because our superpixels have better compactness and boundary adherence. On the other hand, our adaptive superpixel merging method can better encode the multiscale contextual information.

Fig. 13 compares the mean kappa coefficients and the corresponding standard deviation of different methods. It can be found that for different SAR datasets, our method can achieve the largest mean kappa coefficients with a very small standard deviation, indicating that this method is robust to different SAR sensors. In each training, all network parameters of SegNet and Deeplab V3+ need to be initialized randomly, so the kappa coefficients of these two methods both have a large variance. The other four superpixel-level segmentation methods have similar standard deviations.

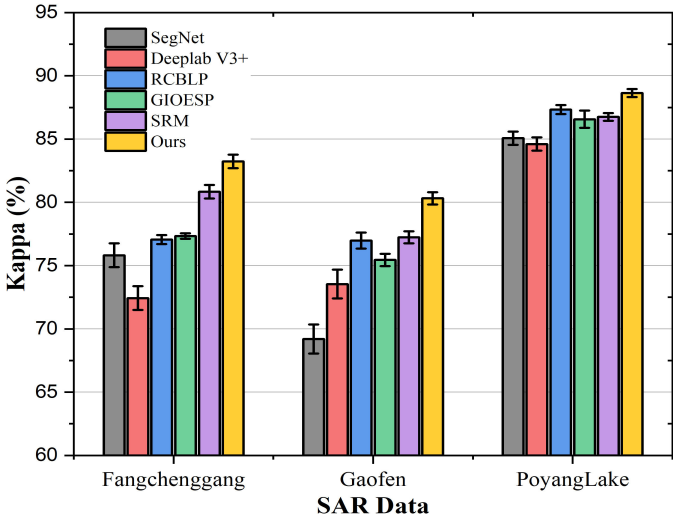


Fig. 13. Mean and standard deviation of kappa coefficients for different methods.

It is worth noting that the Fangchenggang and Gaofen datasets contain more landcovers and have complex backscattering, making accurate segmentation more difficult. While Poyang Lake data is only labeled as two landcovers, the land and water pixels are more distinguishable, so the difficulty of segmentation is relatively low. In theory, our method can modify the shape of superpixels according to the task; hence, it is more suitable for the segmentation of SAR image with complex scenes and weak edges. The experimental results in Fig. 13 are consistent with this analysis, where the proposed method has more significant advantages in Fangchenggang and Gaofen datasets.

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TABLE IV

PER-CLASS ACCURACY, OP (%), OA (%), AND KAPPA COEFFICIENT OF THE OVERALL SEGMENTATION ON POYANGLAKE DATASET

Class	SegNet	Deeplab V3+	GIOESP	RCBLP	SRM	Ours
Water	92.08±0.01	92.68±0.17	96.01±0.33	95.32±0.87	94.35±0.36	**96.71±0.11**
Land	94.51±0.23	93.55±0.24	94.21±0.61	95.61±0.69	**96.05±0.45**	95.68±0.53
OP (%)	93.78±0.44	93.95±0.24	94.68±0.44	95.61±0.66	95.14±0.32	**96.24±0.33**
OA (%)	93.82±0.22	94.69±0.28	95.81±0.53	95.06±0.51	94.65±0.29	**96.61±0.15**
Kappa	85.07±0.53	84.61±0.82	86.57±0.36	87.33±0.68	86.75±0.31	**88.64±0.22**

TABLE V

TIME COSTS (S) OF DIFFERENT METHODS ON THREE REAL DATASETS

Method	Initial superpixel size	SegNet	Deeplab V3+	GIOESP	RCBLP	SRM	Ours
Fangchenggang (1024×1024)	16×16	217.5	268.5	89.3	41.8	38.4	**26.8**
	32×32			27.6	19.3	18.7	**15.1**
Gaofen (512×512)	16×16	99.6	116.4	30.6	15.4	14.9	**10.8**
	32×32			10.5	7.1	6.4	**4.5**
PoyangLake (1024×1024)	16×16	164.3	234.8	88.7	38.9	36.7	**31.5**
	32×32			23.2	18.6	17.6	**14.9**

E. Effect of Initial Superpixel Size on Segmentation Results

For all superpixel-wise segmentation methods, the initial size of the superpixel is one of the most critical parameters, and it determines the segmentation accuracy to a certain extent. So, in this section we compare the results of multiple methods under different superpixel sizes. Fig. 14 depicts the OP and kappa coefficient curves of the proposed methods under different size of superpixels on three datasets, respectively. The horizontal axis represents the different sizes of initial superpixels. Note that "8" in the horizontal axis indicates that the size of initial superpixel is set as 8 × 8. Although the experimental results on different datasets are slightly different, the size of the superpixel has a great impact on our method's performance. The proposed method generally can obtain satisfactory segmentation when the initial size of the superpixel is less than 32 × 32. In particular, when the initial superpixel is approximately 16 × 16, our method can obtain the best kappa coefficients on the Gaofen and PoyangLake datasets. For the Fangchenggang dataset, the ideal initial superpixel size is 20 × 20. When this parameter is greater than 32 × 32, the accuracy of the segmentation results will drop rapidly. Too large superpixels cause that a lot of weak edges are neglected, and some different landcover pixels are divided into the same superpixels. In addition, too small superpixels are also not necessary. This is because the shapes of the small superpixels are more susceptible to speckle geometric distortion and complex backscattering within SAR images. In addition, the smaller size means more superpixels, resulting in high computational complexity in merging phase. Therefore, we set the initial superpixel size in this article to 16 × 16 in the experimental section.

F. Comparison of the Computation Costs

Table V counts the mean running time of various algorithms to segment an image of three SAR datasets when the initial superpixel is 16 × 16 and 32 × 32, respectively. The experiments run on a PC with 32G memory, Intel i7 CPU, Ubuntu 16.04, python2.7, and Pytorch 1.0. For the superpixel-wise method, the running time consists of over-segmentation, superpixel-level feature extraction, and superpixel merging steps. When the dataset and the initial superpixel size are the same, we can get a consistent conclusion by comparing the running time of different methods. For instance, when the image is from Fangchenggang dataset and initial superpixel size is set as 16 × 16, the running times of our method is 26.8 s, whereas the time costs of GIOESP, RCBLP, and SRM are 89.3, 41.8, and 38.4 s, respectively. It can be concluded that our method has the highest computation efficiency. This is because there are only limited number of layers in the our superpixel generation subnetwork, and the following superpixel merging stage only involves calculating a specific power of the similarity matrix, so the entire end-to-end network has relatively fewer computation costs. SegNet and Deeplab V3+ are both pixel-wise encoder–decode structure network, and their network parameters are far more than our method. In particular, the encoder of Deeplab V3+ has 101 layers, resulting in heavy computational burden. Their running time is both more than 200 s, far longer than those of other four superpixel-wise methods. This demonstrates that taking superpixels as processing unit could greatly reduce computational complexity.

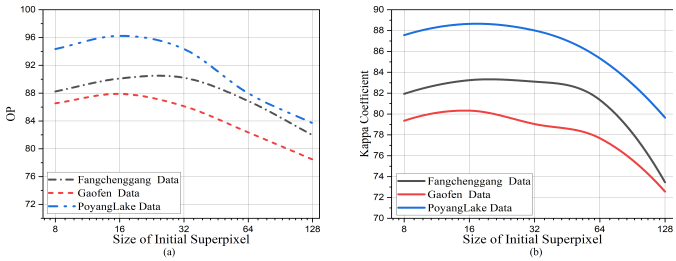


Fig. 14. OP and kappa coefficient curves of proposed method with different superpixel sizes on three datasets: (a) OP curves and (b) kappa curves.

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For the GIOESP method, calculating each two adjacent superpixels takes up a lot of computing resources, so that its running time is basically twice that of the method proposed in this article. The time cost of this method increases sharply with the number of superpixels. The RCBLP method introduces the nearest neighbor graph to reduce the running time of searching minimal edge. Therefore, RCBLP runs faster than GIOESP. The main time consumption of RCBLP comes from the computation of texture pattern similarity measure for each pixel pair. Overall, although our proposed method considers multiple cues during superpixel merging, the method is still highly efficient compared with other state-of-the-art methods.

VI. Conclusion

This article has presented a fast segmentation method for single-polarization SAR images with deep task-specific superpixel merging. First, we introduce a deep network to predict the soft association maps between pixels and pixels, whose elements are statistical likelihood that pixels belong to superpixels. Existing clustering or classification methods cannot direct merge such soft superpixels. Inspired by the concept of the number of walks in graph theory, we present a computationally simple, differentiable, and manual parameters free clustering method named Soft Connection Center. In each iteration, it can intelligently determine the cluster centers and merge the superpixels with the suitable centers according to entries of the similarity matrix. The superpixel generation and merging parts can be implemented under a unified deep network framework. Experiments on different SAR datasets show that our method can gradually adjust the shapes of superpixels according to both the boundaries and landcover classes during training. The proposed model can obtain most of the edges accurately and has fewer confusion. Due to the simple network structure, the segmentation method is computationally efficient and consistently achieve the state-of-the-art performance with good generalizability. In future research, we will further improve this method to make it suitable for polarimetric SAR data.

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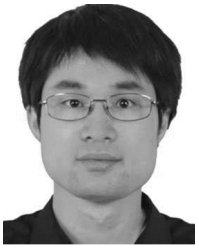
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